



Module	SE101.3 -OOP with Java – 25.3
Lab sheet Number	10
Areas covered	Inheritance + Encapsulation + Polymorphism + Abstraction

Question 1 (Basic Abstract Class)

- Create an abstract class called **Employee** with attributes **employeeId** and **name** and add abstract methods **calculateSalary()** and **displayDetails()**.
- Create subclasses **FullTimeEmployee** and **PartTimeEmployee**.

FullTimeEmployee salary = monthlySalary

PartTimeEmployee salary = hoursWorked × hourlyRate

- In the main () method, create one object from each subclass and display their details and salary.

Question 2

School Management System

- Create an **abstract class** called SchoolMember with the following **private** attributes:

- memberId
- name

Provide:

- A constructor
- Getters and setters
- Abstract methods:
 - displayRole()

- displayDetails()

b) Create an interface called Payable with the following method:

double calculatePayment();

c) Create the following:

Teacher

- Extends SchoolMember
- Implements Payable
- Store:
 - subject
 - monthlySalary

Administrator

- Extends SchoolMember
- Implements Payable
- Store:
 - department
 - monthlySalary

Each subclass should:

- displayRole()
- displayDetails()
- Implement calculatePayment()

d) In the main() method:

- Create one Teacher object and one Administrator object.
- Store them in an array of SchoolMember.
- Use a loop to display each member's role and details.

- Display the payment separately by calling calculatePayment() for each object.

Question 3 - Vehicle Rental System

A vehicle rental company needs a system to manage different types of vehicles.

a) Create a class **Vehicle** with the following private attributes:

- vehicleNumber
- brand
- daily rental rate

Provide a constructor, getters, and setters.

Every vehicle should be able to:

- display its details.
- calculate the rental charge for a given number of days.

b) Create the following, they should **extend** Vehicle class:

c) In the main() method,

- Create one Car object and one MotorBike object.
- Store them in an array of Vehicle.
- Use a loop to display each vehicle's details and rental charge for 5 days.

Question 4 - Hospital Management System

A hospital wants to develop a system to manage its staff members.

a) Create a class **HospitalStaff** with the following private attributes:

- staffId
- name

Provide a constructor, getters, and setters.

Each staff member should:

- display its role according to its type.
- calculate its salary according to its type.
- provide an appropriate medical service.

b) Create an interface called **MedicalService** with the following method:

```
void provideService();
```

c) Create the following, should extend the **HospitalStaff** class:

Doctor

- specialization
- consultationFee

Nurse

- ward
- monthlySalary

d) In the main() method,

- Create one Doctor and one Nurse.
- Store them in an array of HospitalStaff.
- Display the role and salary of each staff member.

Question 5 - Banking System

A bank needs a system to manage different account types.

a) Create a class **BankAccount** with the following private attributes:

- accountNumber
- accountHolderName
- balance

Provide a constructor, getters, and setters.

b) Every bank account should support:

- Depositing money
- Withdrawing money
- Displaying account details

c) Create an interface called **InterestCalculator** with the following method:

double calculateInterest();

d) Create the following, should extend the BankAccount class:

SavingsAccount

- interestRate

FixedDepositAccount

- interestRate
- duration

Each subclass should:

- Provide its own implementation for deposit ()
- Provide its own implementation for withdraw()
- Implement the calculateInterest() method.
- displayAccountDetails().

e) In the main() method,

- Create one SavingsAccount and one FixedDepositAccount.
- Store them in an array of BankAccount.
- Display the account details of each object.
- Display the calculated interest for each account.